

The empirical recurrence for OEIS A317820, recovered from its tail

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Abstract

OEIS A317820 records “Empirical recurrence of order 94” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 240$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 6$, so the recurrence is proved for every $n \geq 100$. Its last coefficient is $c_{94} = 2500 \neq 0$, so no further tail satisfies anything shorter; any order-94 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A317820 is “Number of $n \times 5$ 0..1 arrays with every element unequal to 1, 2 or 3 horizontally, vertically or antidiagonally adjacent elements, with upper left element zero.”. It has offset 1 and begins

3, 160, 356, 2726, 8556, 47374, 179111, 878802, ...

The entry states:

Empirical recurrence of order 94 (see link above)

As of the “Last modified” line on the live entry (Aug 07 2018 22:06:48, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 240$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 240$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 94: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 6$ is the first whose minimal order is exactly the stated 94.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 480$ exact terms from $n = 6$ on, it returns order 94 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{94} c_i a(n-i),$$

c_1	0	c_{25}	−19729620219	c_{49}	−5085367932235	c_{73}	−103226541723
c_2	48	c_{26}	−20254565311	c_{50}	509441252278	c_{74}	−28159127628
c_3	31	c_{27}	66566136280	c_{51}	3095930150713	c_{75}	15150576411
c_4	−1041	c_{28}	9276323215	c_{52}	−319160713185	c_{76}	13472593002
c_5	−1110	c_{29}	−160180856767	c_{53}	533316056966	c_{77}	2923084068
c_6	13737	c_{30}	86345381470	c_{54}	−1020124054716	c_{78}	−5600404059
c_7	18409	c_{31}	276052549628	c_{55}	−3136820996194	c_{79}	−1906480663
c_8	−122857	c_{32}	−355722850922	c_{56}	1761330179942	c_{80}	1429510009
c_9	−183108	c_{33}	−303228546125	c_{57}	3208124190248	c_{81}	467267417
c_{10}	778496	c_{34}	804712561979	c_{58}	−812380221881	c_{82}	−132155231
c_{11}	1170370	c_{35}	58868902720	c_{59}	−1731238894528	c_{83}	−106384073
c_{12}	−3508564	c_{36}	−1194679540135	c_{60}	−861352254197	c_{84}	−22989242
c_{13}	−4669440	c_{37}	568017929529	c_{61}	426277291369	c_{85}	27514444
c_{14}	10382183	c_{38}	1038561642033	c_{62}	1563293067844	c_{86}	6557698
c_{15}	8995114	c_{39}	−1373744978584	c_{63}	229926673415	c_{87}	−4766430
c_{16}	−9648425	c_{40}	31270308758	c_{64}	−1025717069354	c_{88}	−1206479
c_{17}	9989295	c_{41}	1701848832300	c_{65}	−580575368596	c_{89}	264835
c_{18}	−105287834	c_{42}	−1690637075622	c_{66}	243314294474	c_{90}	371423
c_{19}	−77240875	c_{43}	−744398604100	c_{67}	716800000581	c_{91}	7402
c_{20}	826320431	c_{44}	2831070523350	c_{68}	111493961575	c_{92}	−57620
c_{21}	−236179092	c_{45}	−1634330805630	c_{69}	−578671832292	c_{93}	1600
c_{22}	−3618828869	c_{46}	−2454743997097	c_{70}	−126044526416	c_{94}	2500
c_{23}	3739031298	c_{47}	4240494577988	c_{71}	309289184807		
c_{24}	10779006654	c_{48}	837562769667	c_{72}	64276348534		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 100$, and it is the recurrence OEIS A317820 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 6$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{94} - c_1x^{94-1} - \dots - c_{94}$. Its constant term is $-c_{94} = -2500$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 94 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 94, so $\deg q = 94$, and it is monic. It holds from some index t on. From $\max(t, 6)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 94, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 21 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 94, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 2500.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-94 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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